

CH 5440 Multivariate Data Analysis in Process Monitoring and Diagnosis

Assignment 3

1. Principal Component Analysis: A zoologist obtained measurements of the mass (in grams), the snout-vent length (SVL) and hind limb span (HLS) in mm of 25 lizards. **The mean and covariance matrix of the data about the mean** are given by

$$\bar{x} = \begin{bmatrix} 9 \\ 68 \\ 129 \end{bmatrix} \quad S = \begin{bmatrix} 7 & 21 & 34 \\ 21 & 64 & 102 \\ 34 & 102 & 186 \end{bmatrix}$$

(a) The largest eigenvalue of the above covariance matrix is 250.4009. Determine the normalized eigenvector corresponding to this eigenvalue. Also determine the remaining eigenvalues and corresponding mutually orthogonal eigenvectors.

(b) How many principal components should be retained, if at least 95% of the variance in the data has to be captured?

(c) Assuming that there are two linear relationships among the three variables, determine one possible set of these linear relations.

(d) Using the PCA model, determine the scores for a female lizard with the following measurements: mass = 10.1 gms, SVL = 73mm and HLS = 135.5mm.

(e) Using the PCA model, estimate the mass of a lizard whose measured SVL is 73mm

(f) Using the PCA model, estimate the mass of a lizard whose measured SVL is 73mm and measured HLS is 135.5 mm.

Note: This problem has to be solved manually (you can verify the results using a computer code)

2. Yale faces data set is an image data set of 15 subjects that is used for testing facial recognition methods. There are six different grayscale images of each subject with different facial expressions. The resolution of each image is 243 x 320 pixels.

(i) Convert each image into a column vector of size 77760 x 1 by stacking the columns of the digital image one below the other (also known as the vec

operation on a matrix). Using the normal images of each subject as the standard, determine how many of the other images you are able to correctly identify by using Euclidean distance between the image and the standard images.

(ii) Perform a PCA of the normal images (each subject is a sample) and use % variance captured for choosing the number of PCs. Instead of storing the images, the scores of the retained PCs for each standard image can be stored (leading to substantial reduction in storage). Determine how many of the other images you are able to correctly classify based on comparison of the compressed images scores (use Euclidean distance between the score vectors). Vary the number of PCs chosen to find out the optimal number of PCs that gives the best classification accuracy.

3. A flow process is given in Fig. 1. The flow rates of all streams are measured with identical flow meters which have the same accuracies. The covariance matrix computed using 1000 samples from different steady states is given in file cov.mat

(a) How many linear constraints exists among the flow variables for this process?

(b) Determine the eigenvalues of the covariance matrix. Apply hypothesis test for equality of eigenvalues to determine the number of constraints starting from the maximum possible number and graduating decrementing it until 2. Report the degrees of freedom, test statistic, and test criterion in the form of a table. Are you able to determine the correct number of constraints based on the hypothesis test?

(c) Obtain an estimate of the steady state flow constraint matrix using the given covariance matrix of measurements. Compare the estimated constraint matrix with the true constraint matrix using subspace angle.

(d) Based on the estimated constraint matrix, identify all sets of variables that are poor choices of dependent variables. Explain how you identified these sets.

(e) From the estimated constraint matrix obtain the regression matrix relating flows of streams 1, 2 and 3 (chosen as dependent variables) to flows of streams 4, 5 and 6 (chosen as independent variables). Obtain the maximum absolute difference between estimated coefficients of regression and true regression matrix obtained from flow balances.

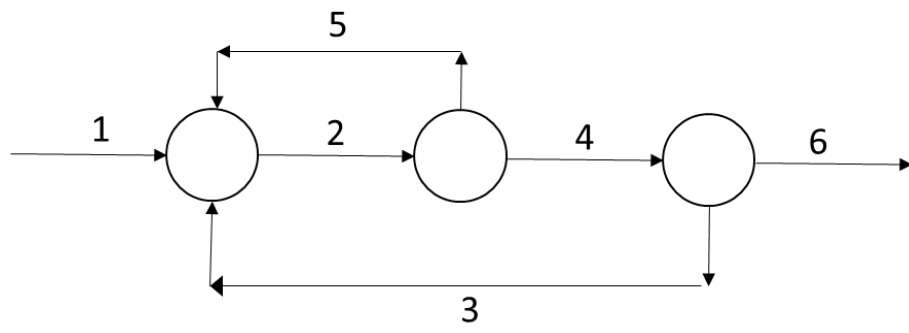


Figure 1